

Transformer

Transformer transforms and changes how the Neural Network like Seq2Seq generate sequential data such as text, voice, or time-series forecast...

Transformer is the backbone of modern LLMs

So the conventional approach in generating sequence of text produce 2 bottlenecks:

- **Sequential dependency:** Each step had to wait for the previous one, making parallelization difficult.
- **Long-range Forgetting:** Information from early in a sequence often degraded by the time it reached later step.

Key Mechanism

- **Self-Attention mechanism:** It allows every token in a sequence to directly attend to every other token, regardless of distance. This solve the long-range dependency problem.
- **Multi-Head Attention (MHA):** Rather than computing a single attention function, the model runs multiple head attention in parallel, each potentially learning to focus on different types of relationship (e.g., Syntactic vs Semantic)
- **Positional Encoding: Since attention** has no inherent notion of order, positional encodings are added to input embeddings to inject sequence positional information, originally using a fixed sinusoidal function.
- **Encoder-Decoder Architecture:**
 - **An encoder:** to build a contextual representation of the input
 - **A decoder:** to generate output tokens autoregressively (generate one element at a time)